Andrés Martínez Silva

Robotics Software Engineer

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Profile

Robotics software engineer with 3+ years of R&D experience in European and international research projects (H2020, Honda Research Institute Japan), spanning three areas: state estimation and sensor fusion (multi-robot localization fusing UWB, Radar and IMU with factor graphs; first-author paper at ICRA 2026), control (UAV guidance and autonomous landing validated in real flights, model-based and MPC control of robot actuators) and robotics simulation (creator of UWBPX4Sim, an open-source PX4/Gazebo simulator presented at ICRA 2026 and ROSCon Spain 2026). Strong academic background in SLAM, computer vision and reinforcement learning. Experienced in taking algorithms from C++/Python and ROS 2 implementation through Software-in-the-Loop simulation to real-world validation.

Skills

Languages: C++, Python, MATLAB/Simulink
Frameworks: ROS 2, Nav2, PX4, Gazebo, Webots, Unity, Ceres Solver, TensorFlow
Tools: Git, Linux, Docker, Kubernetes, RViz
Areas: Localization, SLAM, Sensor fusion, Factor graph / pose-graph optimization, Control systems (PID, MPC, system identification), UAV guidance and landing, U-space/UTM, Software-in-the-Loop simulation, Deep Learning, Ultra-Wideband, Radar

Experience

Full-time family caregiving Mar 2026 – Present

- Continued developing UWBPX4Sim and presented it at an ICRA 2026 workshop and ROSCon Spain 2026.

Service Robotics Lab, Universidad Pablo de Olavide Apr 2025 – Mar 2026
Research Engineer *Seville, Spain*

- Developed a UGV-UAV relative localization system fusing Ultra-Wideband ranging, Radar ego-motion, IMU and wheel odometry through nonlinear trilateration and pose-graph optimization (ROS 2, C++, Ceres), outperforming state-of-the-art closed-form methods.
- Validated the system in Software-in-the-Loop simulation and on a real-world dataset (~10 min of simultaneous UAV-UGV motion with UWB, Radar and odometry); released the code and data as an open benchmark on Harvard Dataverse. First-author paper accepted at ICRA 2026 (arXiv).
- Built UWBPX4Sim, the open-source multi-robot UWB simulator used in this work (see Open-Source Projects).
- Developed a Webots plugin for the HuNavSim human navigation simulator, contributing to HuNavSim 2.0 (ROBOT 2025, third author).
- Lead maintainer of the onboard ROS software stack of Haru, a tabletop social robot in a project led by Honda Research Institute Japan: refactored the legacy codebase into modular packages, integrated a new U2D2 controller board (replacing ESP32), raising neck control rate to 100 Hz.
- Upgraded actuator control from open-loop to closed-loop: identified joint dynamics from real data (MATLAB/Simulink), packaged the identified model as a MATLAB library and Unity plugin for simulation, and implemented PID with feedforward and Model Predictive Control for smooth trajectory tracking.
- Shipped 4 production releases in 4 months through 30+ pull requests across 5 repositories (web UI for robot and Docker image management, ROS 1 compatibility, CI with GitHub Actions, pre-shipment stress tests for the manufacturer) and prepared and supported 10+ live demos with international partners.

FADA-CATEC Mar 2022 – Aug 2024
R&D Engineer, Autonomous Systems Unit *Seville, Spain*

- Developed a vision-based autonomous landing module for urban parcel delivery, following EASA vertiport specifications for safe maneuvers, in the DELOREAN project (H2020/EUSPA, 8-partner consortium including Airbus, Correos and Eurocontrol).
- Implemented the module in Python and ROS, tested it in Software-in-the-Loop simulation and validated it in real-world flight experiments, including the project's public demonstration with Correos (Utrera, Spain, 2022), where it achieved reliable landings in strong wind conditions.
- Developed a 5D guidance system (3D position + yaw + time) for unmanned aircraft using the carrot-chasing technique, implemented in C++ and ROS.
- Integrated the guidance system into the U-Space environment within AGORA, a joint Airbus–CATEC research unit on unmanned traffic management.
- Built C++/Python tools for UAV manipulation-platform control, RTK/Total Station frame alignment, RViz-based trajectory annotation and cloud database integration in AERIAL-CORE (H2020, 18 European partners), on aerial robots

for power-line inspection and maintenance.
- First-author paper on autonomous vertiport landing presented at ROBOT2022.

Open-Source Projects

UWBPX4Sim · github.com/amartinezsilva/UWBPX4Sim
Ready-to-use UWB simulator for multi-robot relative localization · ROS 2 Jazzy, Gazebo Harmonic, PX4 SITL, C++, Python, Docker
- Custom Gazebo plugin generating realistic UWB ranges modeled after real data: distance-dependent LOS degradation, NLOS noise and dropout from obstacle thickness along the line of sight, and per-pair bias.
- Configurable fleets of PX4 UAVs and UGVs from a single YAML layout, with automated model and ROS 2 bridge generation, offboard trajectory-tracking nodes and synthetic odometry drift.
- One-command setup script, browser-based configuration GUI and Docker image for reproducible experiments.
- Presented at the ICRA 2026 Workshop on Robot Meets GNSS and Ranging (Vienna) and at ROSCon Spain 2026 (Valencia).

HuNavSim Webots wrapper · github.com/robotics-upo/hunav_webots_wrapper
ROS 2 Humble, Webots, C++, Nav2
- ROS 2 plugin and world generator that integrate the HuNavSim human-agent simulator into Webots, with office, factory and hall scenarios for human-aware navigation of a TIAGo robot.

Publications & Talks

- **A. Martínez-Silva**, D. Alejo, L. Merino, F. Caballero. *Radio-based Multi-Robot Odometry and Relative Localization.* IEEE International Conference on Robotics and Automation (ICRA), 2026. arXiv:2509.26558
- **A. Martínez-Silva** et al. *UWBPX4Sim: Ultra-Wideband simulator for multi-robot applications.* ICRA 2026 Workshop on Robot Meets GNSS and Ranging for Seamless Autonomy, Vienna, 2026. Also presented as a talk at ROSCon Spain 2026, Valencia.
- **A. Martínez**, P. J. Sánchez-Cuevas, M. A. Trujillo, A. Viguria. *Autonomous Landing Strategy for Vertiports: A Preliminary Case Study*. ROBOT2022: Fifth Iberian Robotics Conference, Springer LNNS vol. 590, pp. 389–400, 2023. DOI: 10.1007/978-3-031-21062-4_32
- M. Escudero-Jiménez, N. Pérez-Higueras, **A. Martínez-Silva**, F. Caballero, L. Merino. *HuNavSim 2.0: An Enhanced Human Navigation Simulator for Human-Aware Robot Navigation.* ROBOT 2025: Advances in Robotics, Springer. DOI: 10.1007/978-3-032-30156-7_25

Education

Universidad de Zaragoza 2023 – 2025
MEng in Robotics, Graphics and Computer Vision
- Completed alongside full-time professional work · Final grade: 8.88/10 · 90 ECTS
- Thesis: Radio-based Multi-robot Odometry and Localization
- Relevant coursework: SLAM, Computer Vision, Reinforcement Learning, Deep Learning

Universidad de Sevilla 2021 – 2022
MSc in Electronics, Robotics and Automation
- Completed alongside full-time professional work · Final grade: 8.7/10 · 60 ECTS
- Thesis: Autonomous landing strategy for vertiports: preliminary case study and UAD use case (grade: 10/10)

Universidad de Sevilla 2015 – 2020
BEng in Electronic, Robotics and Mechatronic Engineering (Robotics and Automation)
- Final grade: 7.84/10 · 240 ECTS
- Thesis: Fundamentals of Design and FMG Control Interface for a Hand Prosthesis Based on Grip Generation (grade: 10/10)

Certifications & Additional Training

- **Robotics: Aerial Robotics** · University of Pennsylvania (Coursera) · 2021 · Certificate
- **Deep Learning Specialization** · DeepLearning.AI (Coursera) · 2020 · Certificate
- **Clean Code** · Academind (Udemy) · 2021 · Certificate
- **Docker & Kubernetes: The Practical Guide** · Academind (Udemy) · 2021 · Certificate
- Short courses: Summer Course in Mechanobiology (UNIA, 2020); Campus of Excellence in Robotics (URV, 2015)

Languages

Spanish: Native · **English:** C1 · **French:** A2/B1